

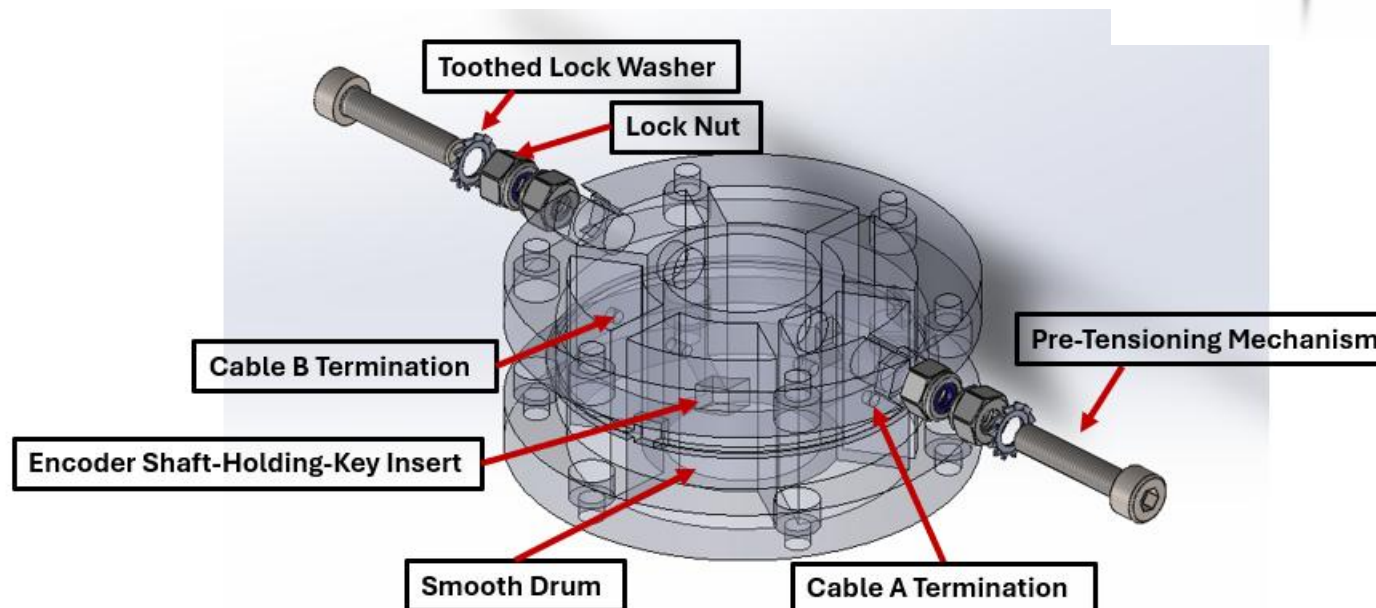
END-TO-END CONTROL SYSTEMS & MECHATRONICS DESIGN (CABLE-DRIVEN MANIPULATOR)

2025 MAY- 2026 MAY

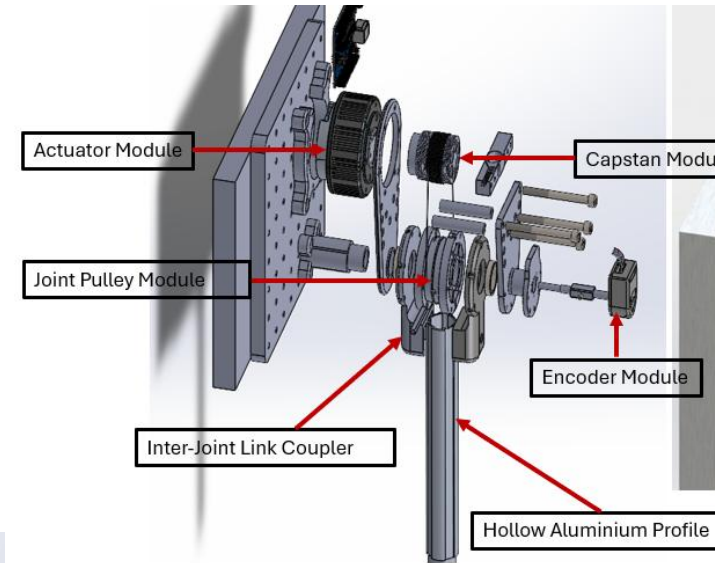
NTU ADVANCED ROBOTICS RESEARCH LAB X SCHAFFLER ADVANCED RESEARCH HUB

(SINGLE JOINT CONTROL VALIDATION)

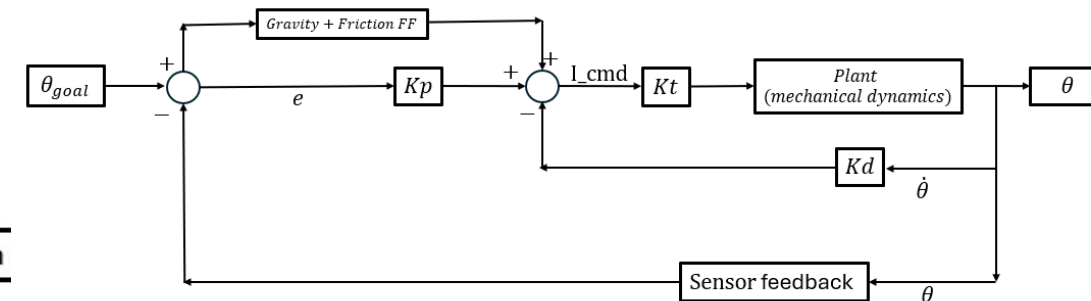
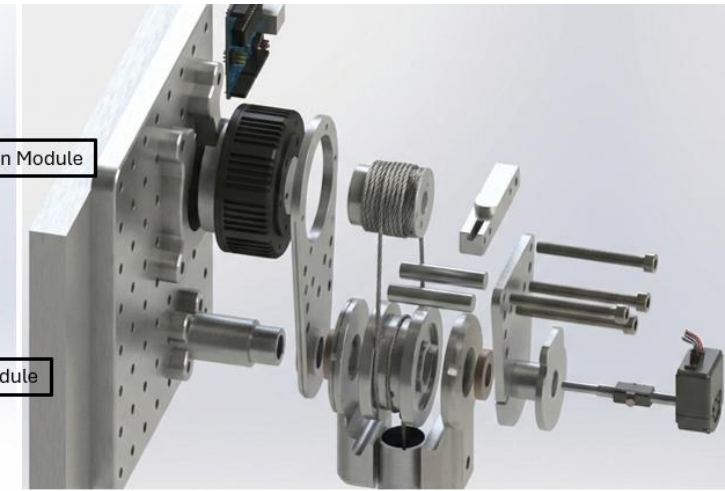
- 10+ Design iterations to optimize cable routing and joint efficiency.
- Real-time CAN-Bus network control for motors
- Implementation of **closed-loop impedance control + gravity compensation feed forward**
- Cable-driven joint accuracy of **>95%** (read implementing joint encoder)
- Full mathematical modelling of the cable-driven joint



SolidWorks View



Rendered View

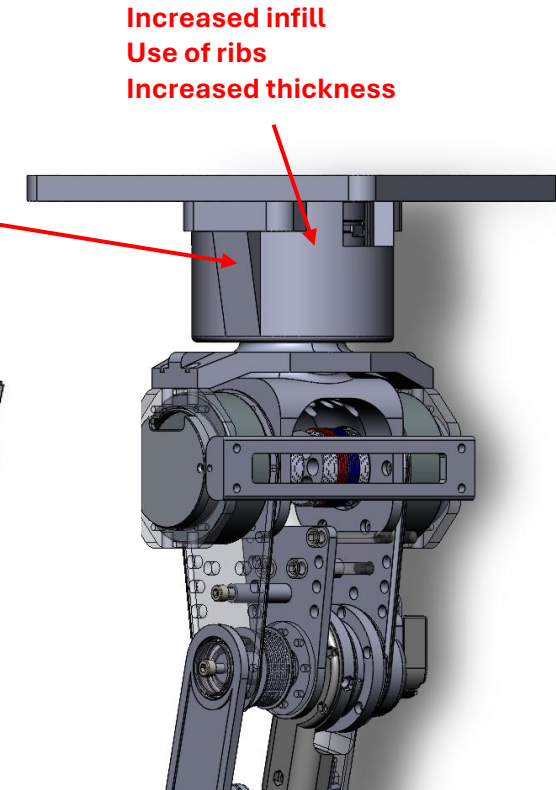
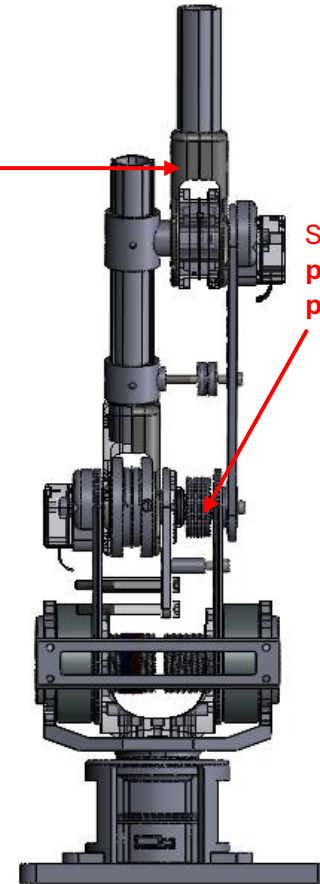
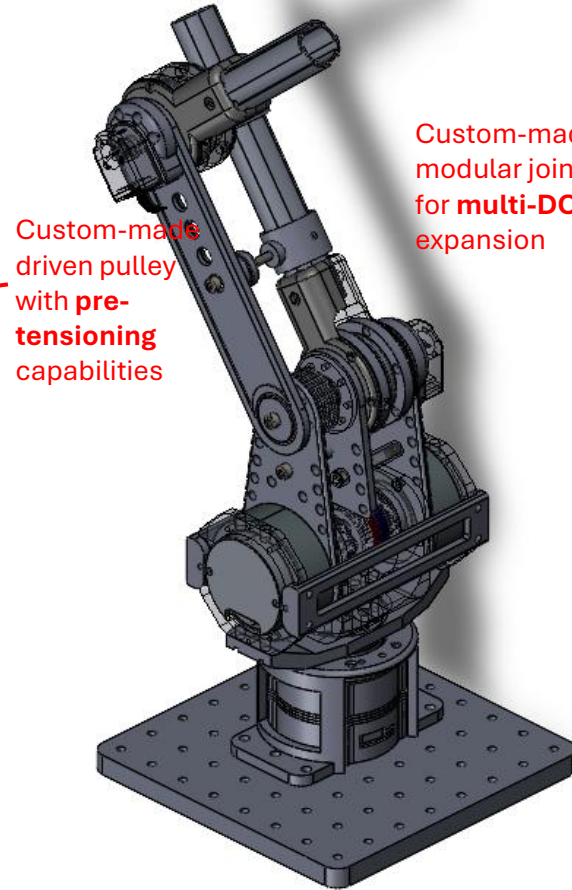
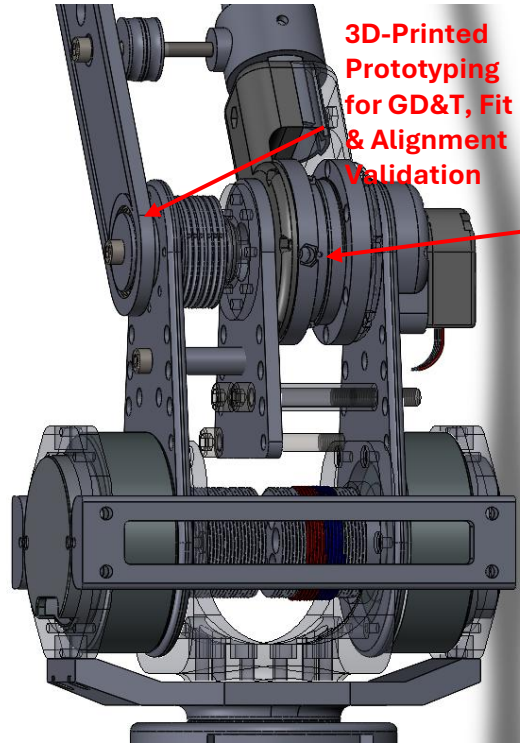


(full detailed metrics confidential)

EXPANSION INTO MULTIPLE-DOF TORQUE-CONTROLLED CABLE-DRIVEN MANIPULATOR

NTU ADVANCED ROBOTICS RESEARCH LAB X SCHAFFLER ADVANCED RESEARCH HUB

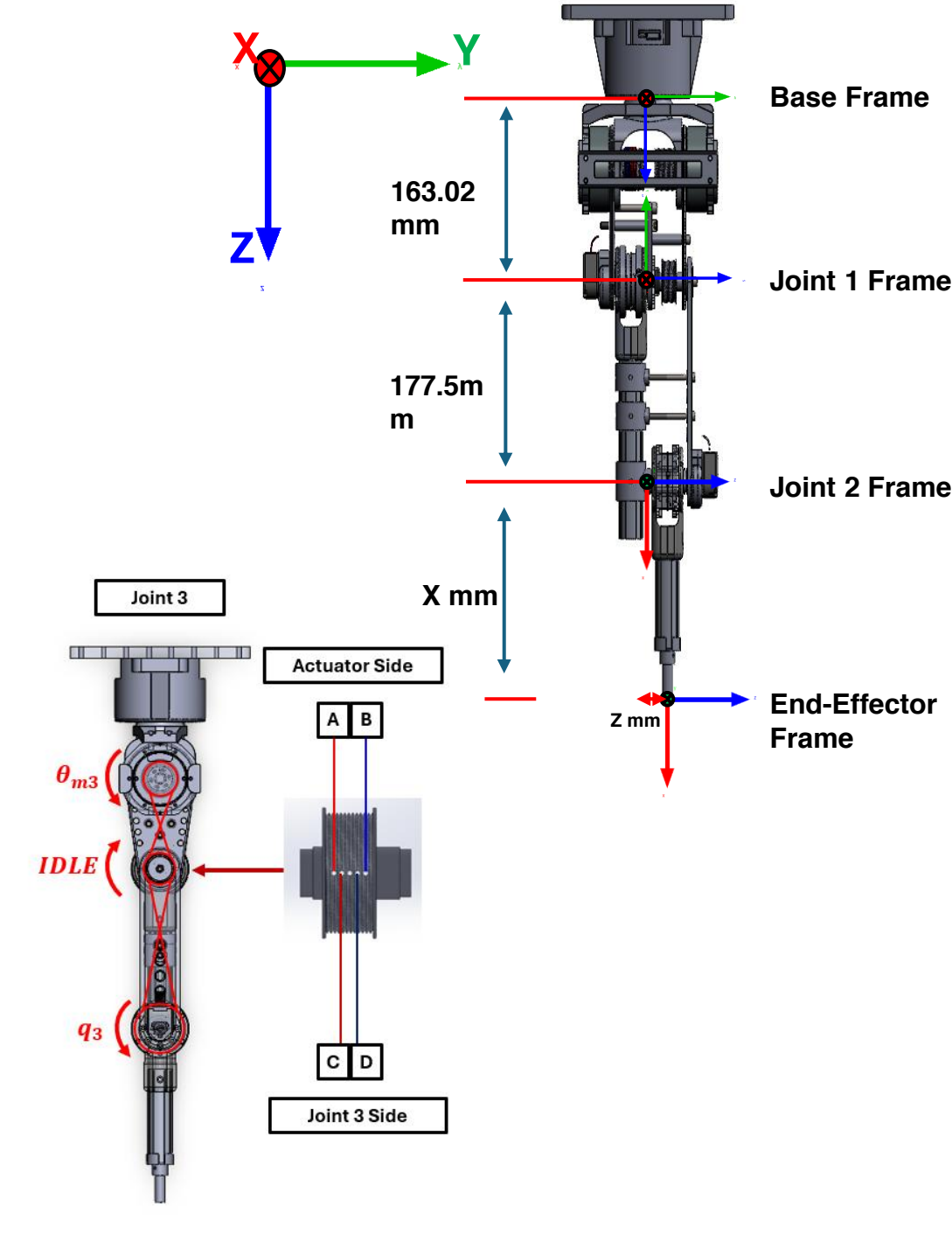
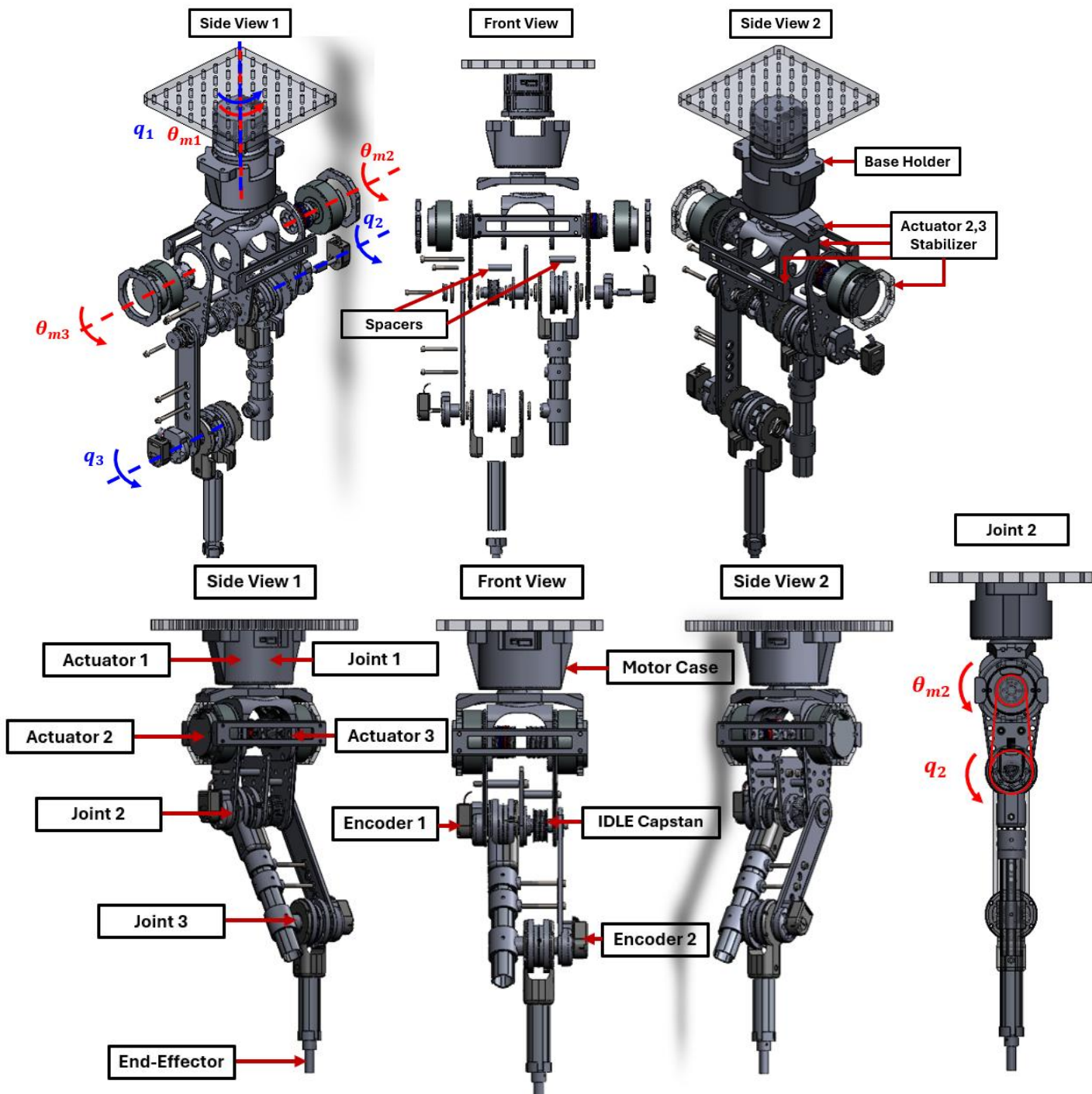
Goal: To perform precise insertion tasks utilizing torque control.

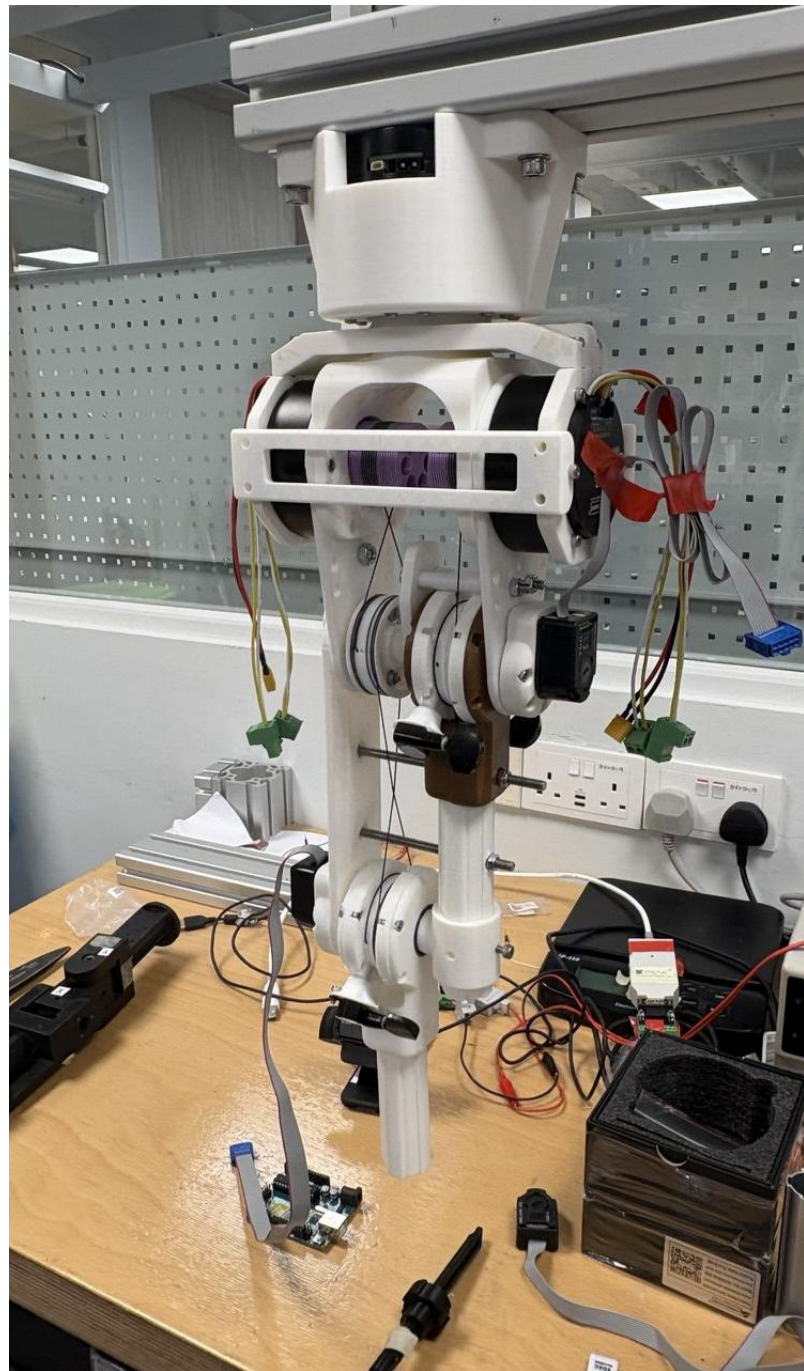
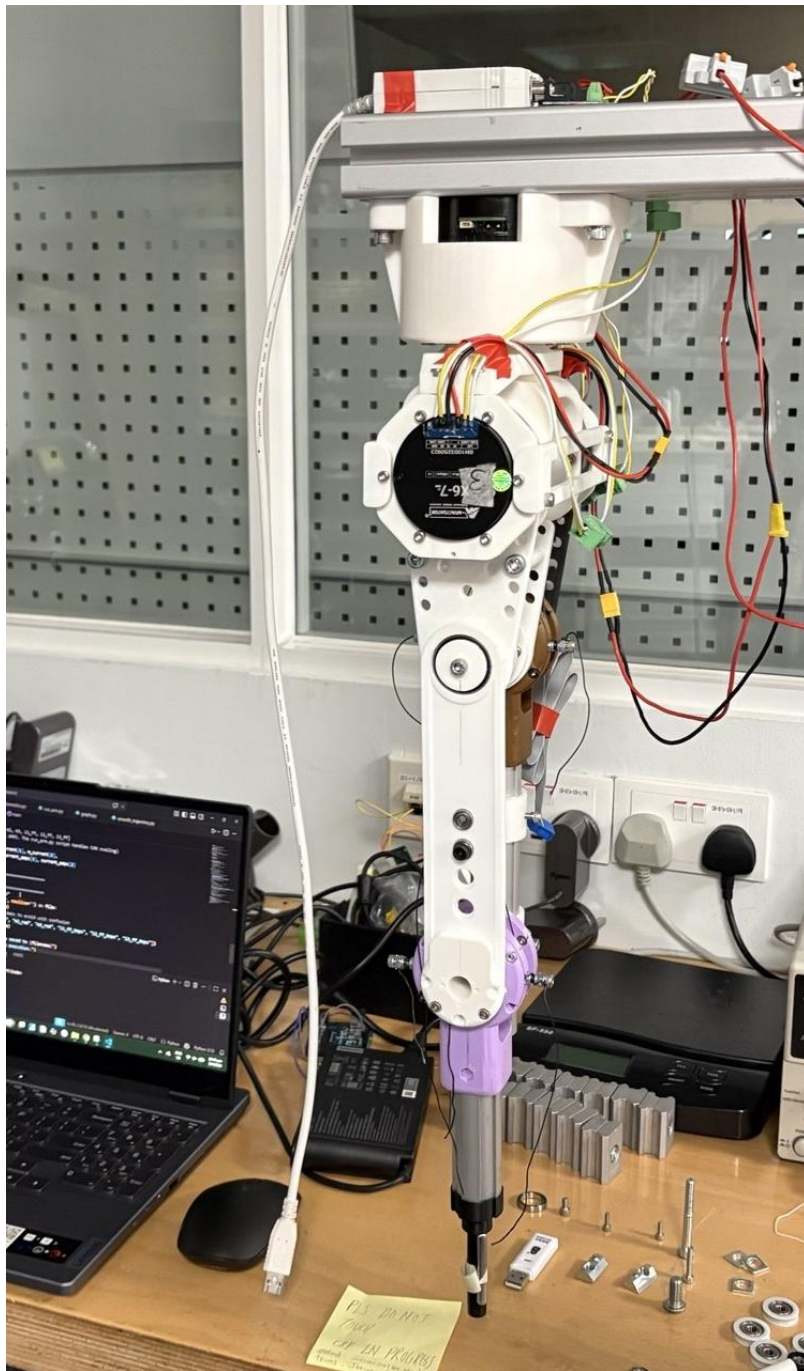
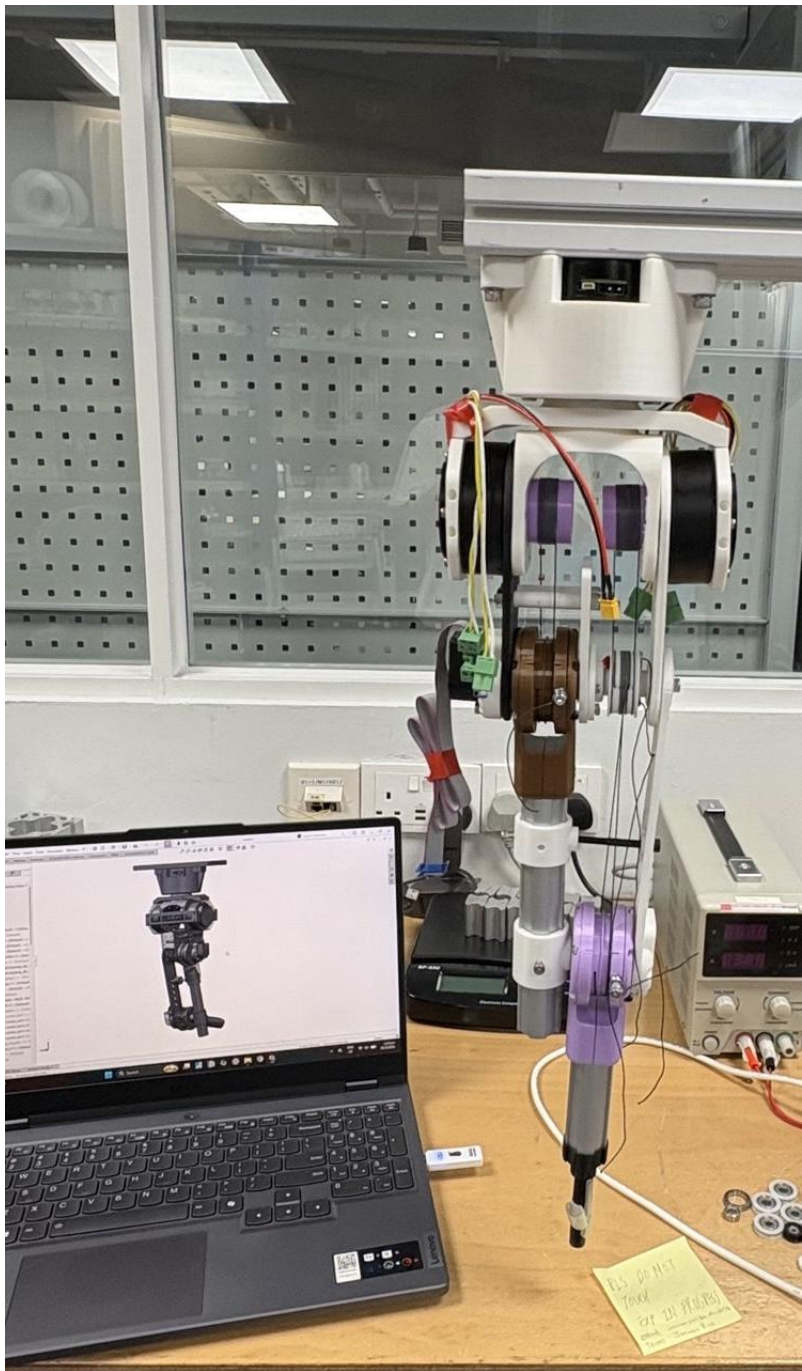


Complete mathematical modelling of the robotic system:

- Forward & Inverse Kinematics
- Dynamic Control: Gravity Compensation + Impedance Control
- Quintic Trajectory Planning

(full detailed metrics confidential)





Hyundai Motor Group Innovation Center (HMGICS)

Robotics Engineering Internship

- **Designed passive mechanical alignment mechanisms for complex fastener geometries** used in a robotic assembly line, achieving **pose consistency** without vision or active sensing.
- **Developed & enhanced robotic tooling and fixtures:** gripper fingers, alignment jigs, and sensor couplers, using SolidWorks. Validated designs through stress analysis and rapid prototyping cycles.
- **Integrated custom mechanical subsystems** into **ABB robotic automation cells**, supporting **pick-and-place** operations through RAPID programming, simulation, and system-level testing.
- **Built experimental test rigs using industrial robot arms** to evaluate material durability and fixture performance, enabling data-driven mechanical design refinement.
- **Collaborated with R&D teams** to transition robotic assembly tooling from prototype to production, contributing to automation improvements targeting major operational cost reduction.
- **Explored & validated robotic motion planning and mechanical designs** across simulation and physical hardware environments.

(detailed metrics & design asset confidential)

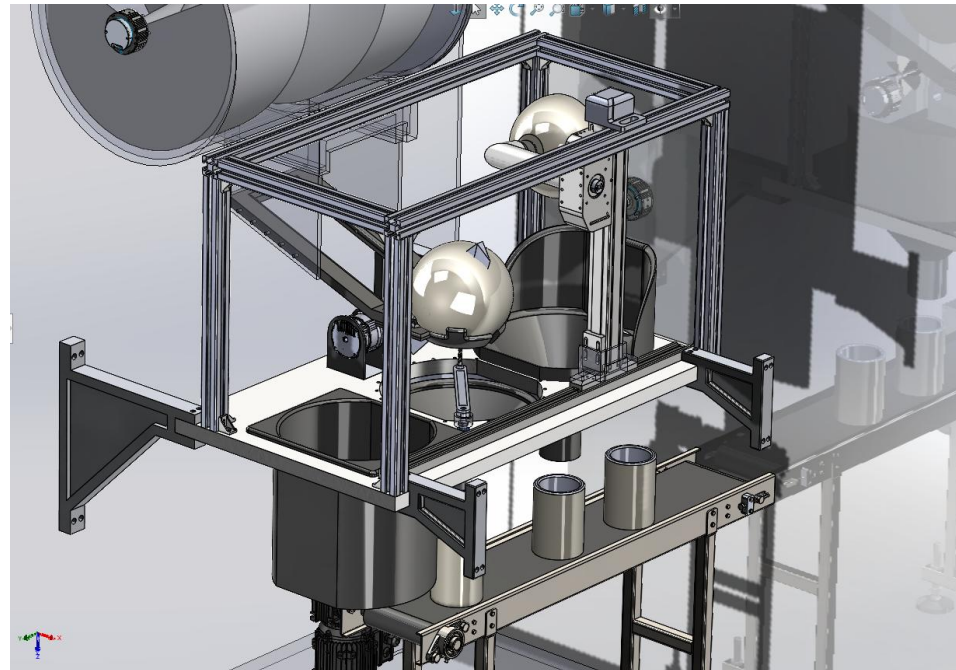
End-To-End Conceptual Design of Automated Coconut Vending Machine

SolidWorks View

1. Automatically load coconut into the machine
2. Empty the juice inside the container
3. Open the coconut
4. Carve out the flesh into the same container

Performed Analysis:

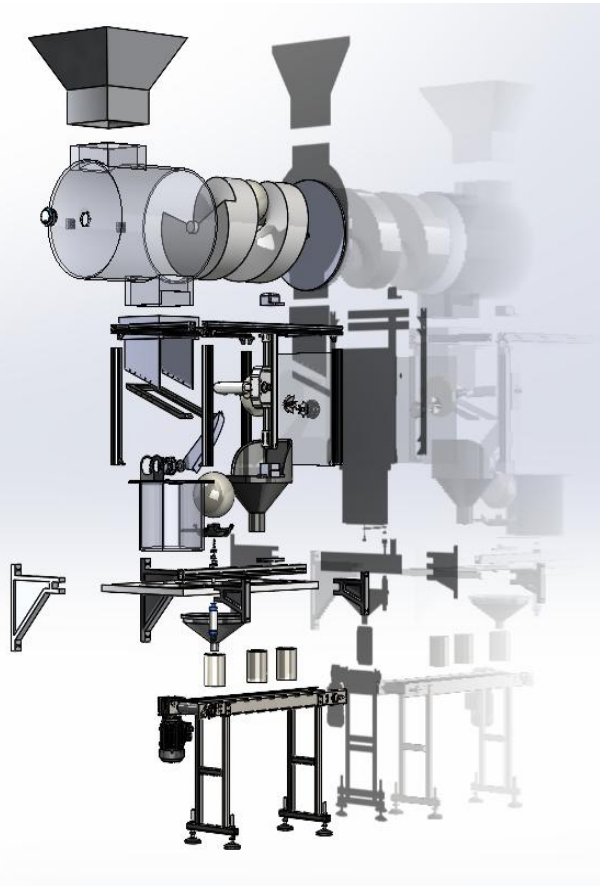
- NC machining tool paths simulation
- Dynamic motion analysis of the machine
- Engineering drawing of parts to be manufactured



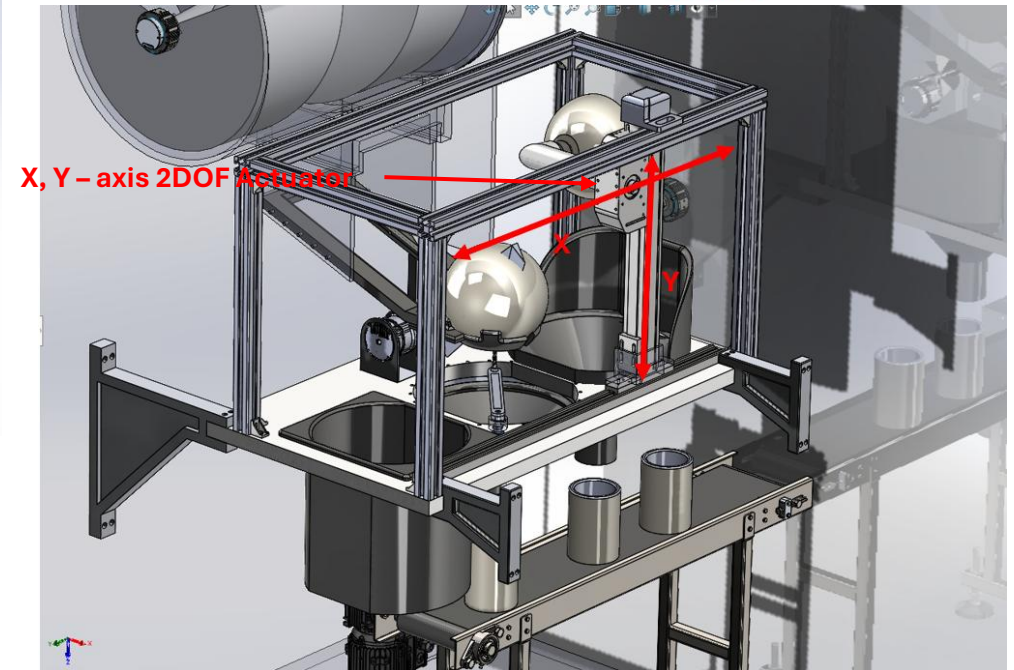
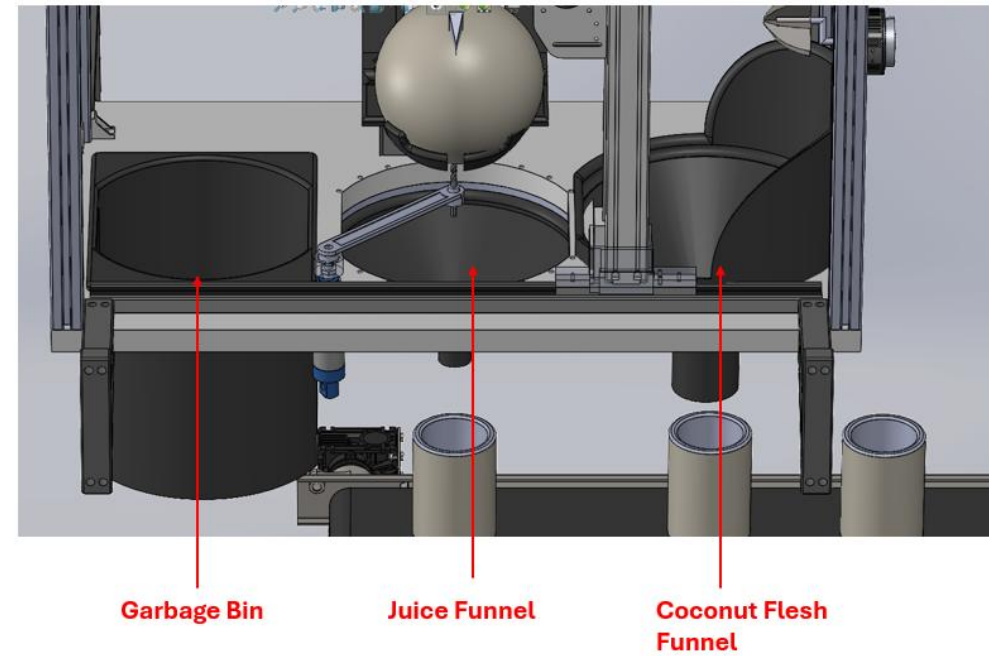
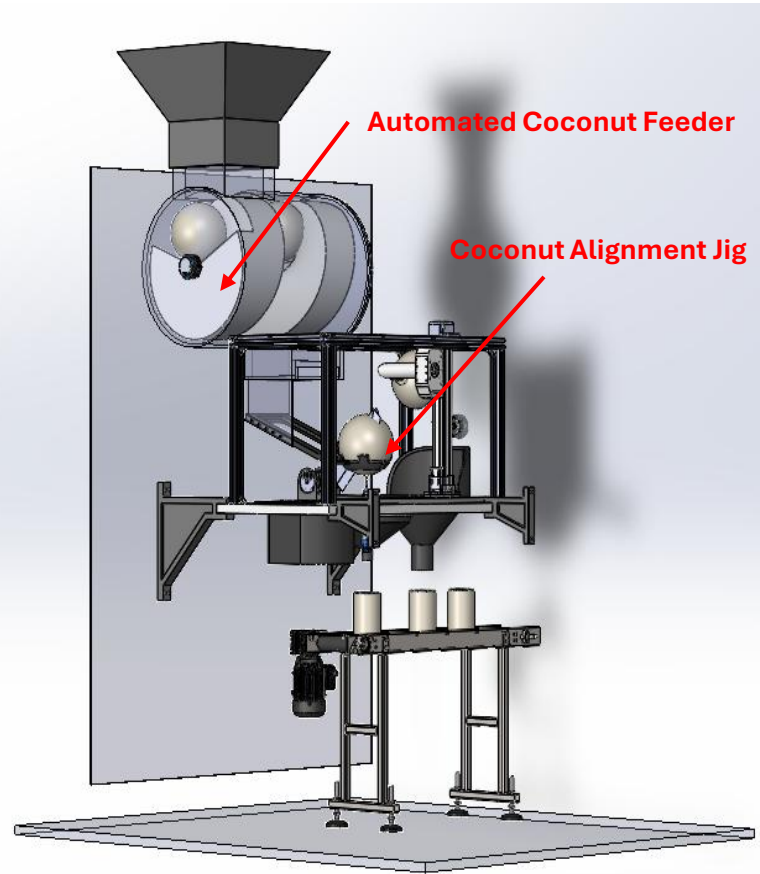
Rendered View



Exploded View



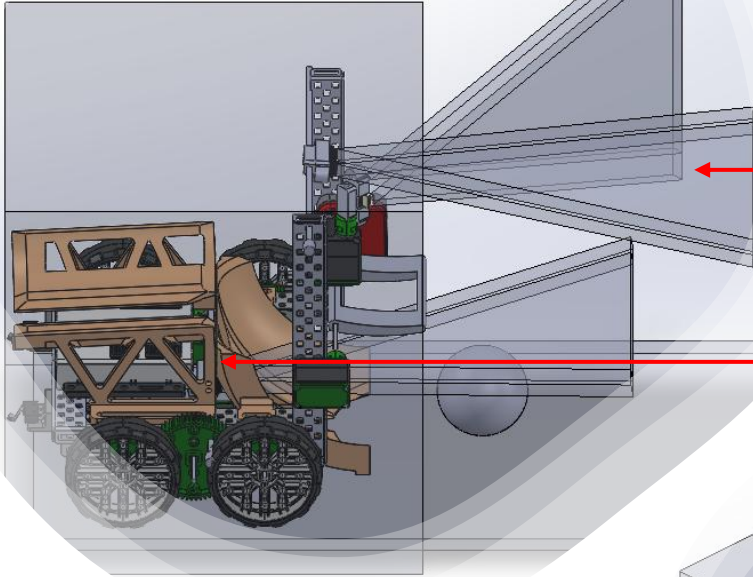
Assembly View



Automated Rapid Ball Picking Robot

2026

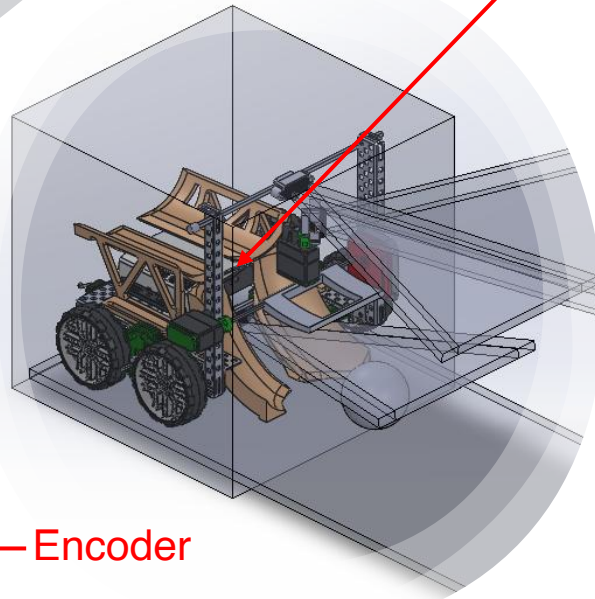
NTU MECHATRONICS DESIGN & ROBOTICS COMPETITION



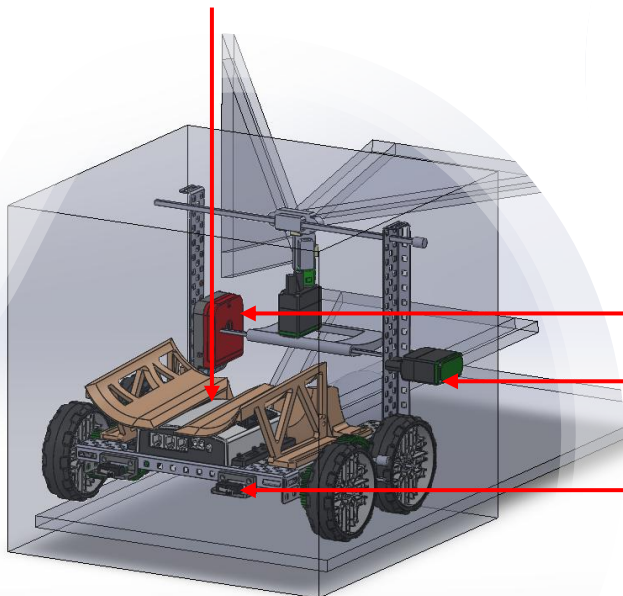
FOV design to ensure no interference with vision

Lowered COM Location (structurally robust)

Complete Internal Cable Management



Ball Guidance Rail



Encoder

Motor

Limit Switches for Obstruction Detection

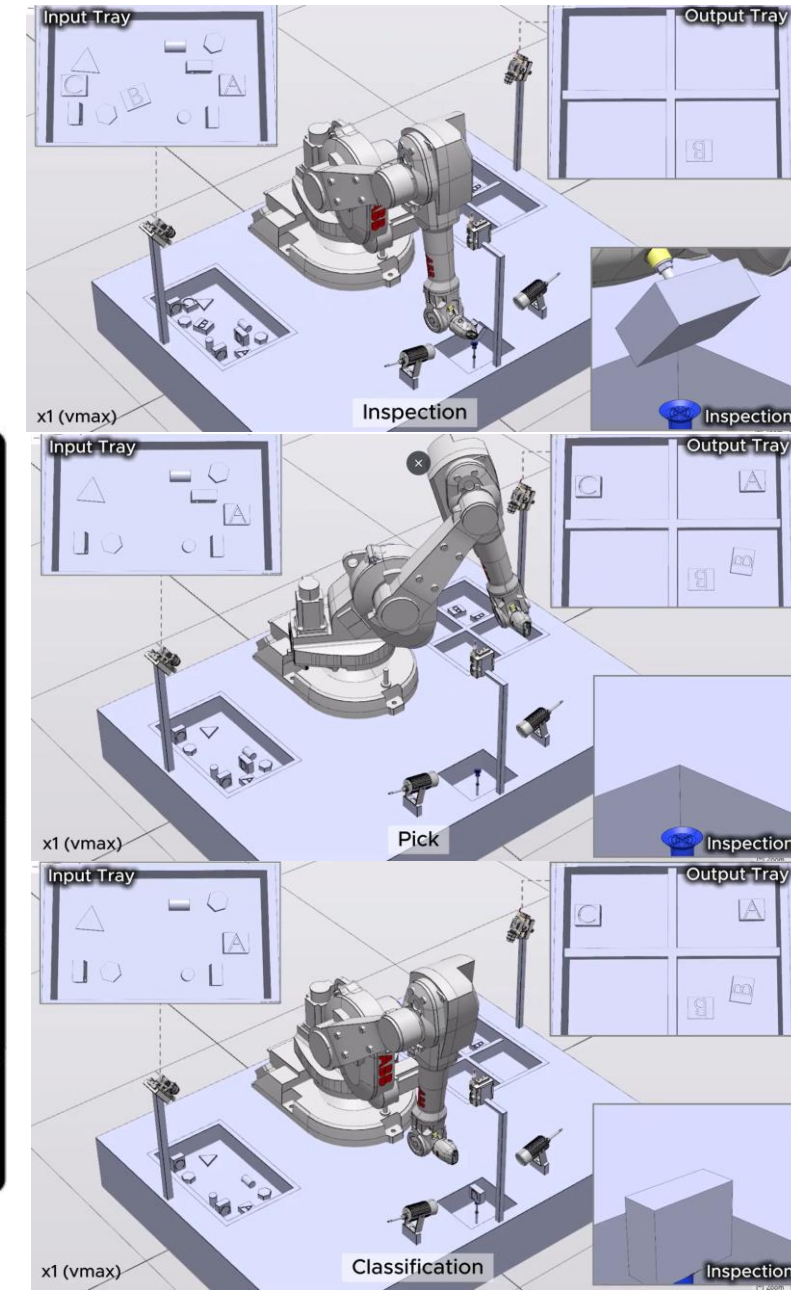
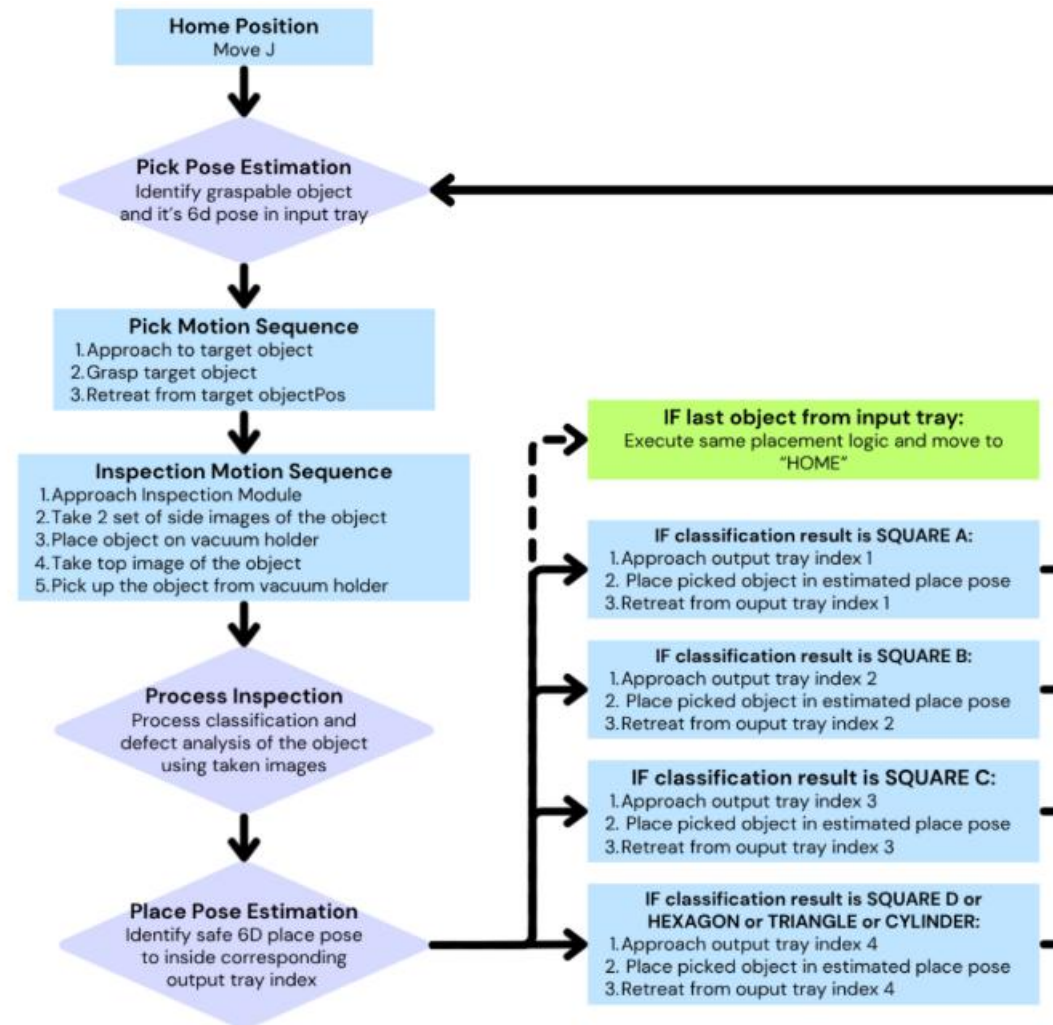
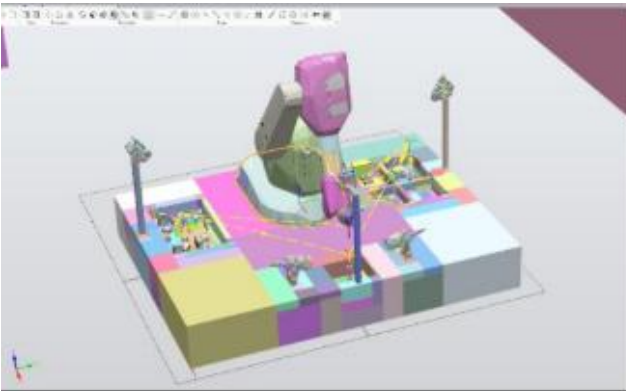
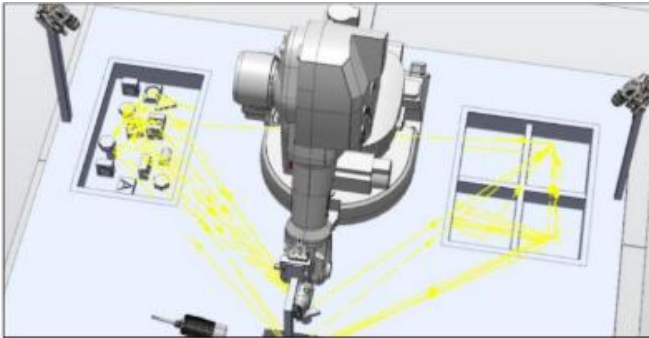
Design Considerations

- Simplicity (removal of redundancy & unnecessary complications)
- Goal Driven
- Compactness
- Robustness
- Cost efficient
- Time efficient (Suitable for rapid prototyping and ease of adjustments according to feedback)
- Aesthetics (Consideration of symmetry, golden ratios, and sleekness of the design)

ABB RobotStudio Path Planning Demonstration

Unstructured item sorting, generating optimized RAPID routines to reduce cycle time.

Designed pickable objects and environment tooling in CAD.



NTU MECATRON Team Autonomous Underwater Robot

2024-2025

SINGAPORE AUV CHALLENGE COMPETITION 1ST PLACE

1. Mechanical Design

- Mechanical Design of the Underwater Robot Shell
- Focused on waterproofing methods + convenience for assembly and disassembly

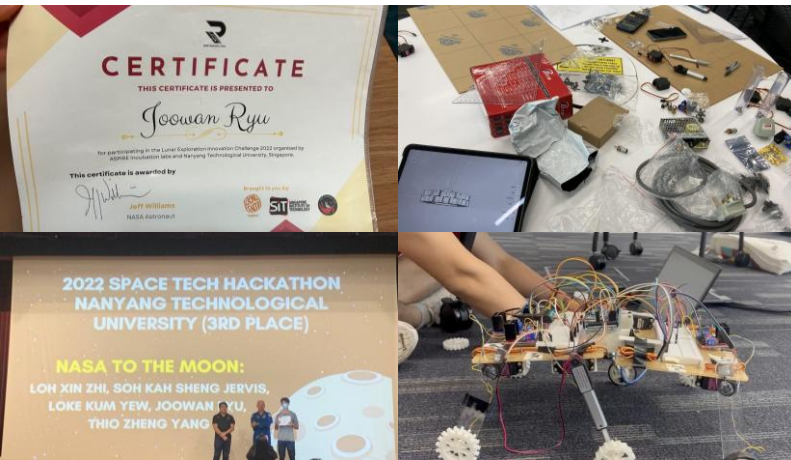
2. Workshop Demonstration

- Designed and fabricated a **modular underwater robot platform** optimized for educational workshops.
- Helped engineer the mechanical structure for **rapid assembly and disassembly**, allowing students to gain hands-on experience with actuation, waterproofing, and mechatronic integration.

Space Hackathon

2022

3rd PLACE / 30 + TEAMS AWARDED BY NASA JEFFREY WILLIAMS

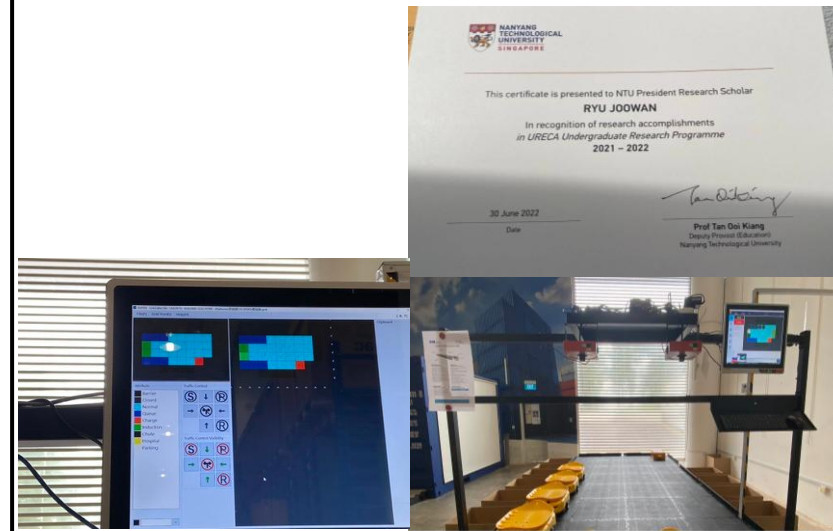


- Designed and prototyped a lunar rover prototype capable of overcoming crater slopes up to 45° within an \$800 development budget.

- Engineered mechanical architecture and traction system to maintain stability on steep terrain.

URECA Research

2022



- Research on the path planning of automated ground vehicles (AGV)
- Use of gazebo, URDF modeling, ROS1 to plan AGV trajectories
- Visit to HAI Robotics & SIT to explore and learn their approach on AGV path planning